CSCI 432/532, Spring 2026

Final Project

The final project for this course is to design and deliver a lecture covering an algorithms topic that we did not cover this semester. In groups of 3, you will select a topic, use outside resources (I can provide some suggestions) to learn about it, and prepare a 50 minute lecture to teach the class about the topic in the context of what we have learned in this course.

1 Project Schedule & Deliverables

- **Friday, March 27, by 10am: group formation**. No deliverable, but make sure I know your group by this date. Every group must have exactly 3 members: one graduate student and two undergraduates.
- **Friday, April 3, by 5pm: topic selection**: Submit a short writeup on Canvas (only one group member will need to submit) giving a *formal description of the problem or topic* and at least two quality resources (textbooks, collections of lecture videos, etc.) that you will use to prepare your lecture.
- **Friday, April 15, in class: teaser presentation**: Give a 10 minute presentation to the class introducing your topic: the basic problem definition, why you chose it, and what you still are working to understand about it. And get us excited to learn more about it!
- **April 29–May 4 in class and finals period**: Deliver a 50 minute lecture lecture to the class on your topic. You must also prepare a handout for the class providing any formal definitions, theorems, and proofs that you cover in your lecture.

2 Grading

Your total project grade will be based on the following. Note that 85 points are earned with the entire group, and 20 points are earned individually by participating and providing feedback on other groups' presentations.

- 0 points: group formation.
- 5 points: topic selection deliverable.
- 10 points: teaser presentation.
- 10 points: handout for final presentation.
- 55 points: final presentation.
- 5 points (individual): self evaluation and evaluation of group members.
- 15 points (individual): audience participation during and providing feedback on other groups' presentations. (3 points teaser presentation, 3 points draft presentation, 3 points each final presentation of other groups).

2.1 Rubrics

Teaser presentation. 10 points =

- 4 points: demonstrates understanding of the problem or topic
- 2 points: all project members participate in the presentation
- 2 points: provide some motivation for the class to learn about the topic
- 2 points: presentation is clear and professional

3 Potential Topics

Here are some topics to consider. Feel free to choose a topic outside of this list. Note that some topics may be too large for a single lecture, so you may need to focus on a specific aspect of the topic. Additionally, some topiocs may require more background than we have covered in class, but they may still be worth taking on if you are excited about them! I am happy to advise on the choice of a topic.

If I know of book chapters or lecture videos that cover a topic, I list them here and point to the specific chapter in the list of topics; you are not limited to these resources, however.

CLRS Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein,jj Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, 3rd ed., MIT Press, 2009.

JeffE Jeff Erickson, *Algorithms*, 2024. Available online at https://jeffe.cs.illinois.edu/teaching/algorithms/.

Avi Avi Wigderson, *Mathematics and Computation: A Theory Revolutionizing Technology and Science*, Princeton University Press, 2019. Available online at https://www.math.ias.edu/avi/book.

KT Jon Kleinberg and Éva Tardos, *Algorithm Design*. Addison-Wesley, 2006.

FGC Virginia Vassilevska Williams, *On some fine-grained questions in algorithms and complexityy*, lecture notes, 2024. Available online at people.csail.mit.edu/virgi/eccentri.pdf.

3.1 Topics

- Advanced computer science theory topics: approximation algorithms [CLRS Ch. 35; KT Ch. 11], parameterized complexity and fixed paramter tractability [KT Ch. 10], fine-grained complexity [FGC], proof complexity [Avi Ch. 6], space complexity [Avi Ch. 14; KT Ch. 9], attacks on P vs. NP [Avi Ch. 5], etc.

- Typical advanced algorithms topics that we don't get to: network flows [KT Ch. 7; CLRS Ch. 29; JeffE Ch. 10-11], linear programming [CLRS Ch. 29; JeffE Ch. H-I], randomized algorithms [KT Ch. 13; CLRS Ch 5; Avi Ch. 7], online algorithms [CLRS Ch. 29], parallel algorithms [CLRS Ch. 25], matrix operations [CLRS Ch. 28], etc.

- Advanced data structures: range queries, hashing and sketching, bloom filters, tries and suffix trees, etc.

- Algorithmic perspectives on machine learning: computational learning theory [Avi Ch. 17], local search [KT Ch. 12], back propogation and gradient descent [CLRS Ch. 33], etc.
- Quantum speedups for machine learning algorithms (quantum annealing, HHL algorithm, quantum kernel method, quantum PCA, etc.)
- More advanced cryptography topics: zero-knowledge proofs [Avi Ch. 10], blockchains, etc.
- Practical solutions to NP-hard problems: integer linear programming, SAT solvers, TSP solvers, etc.